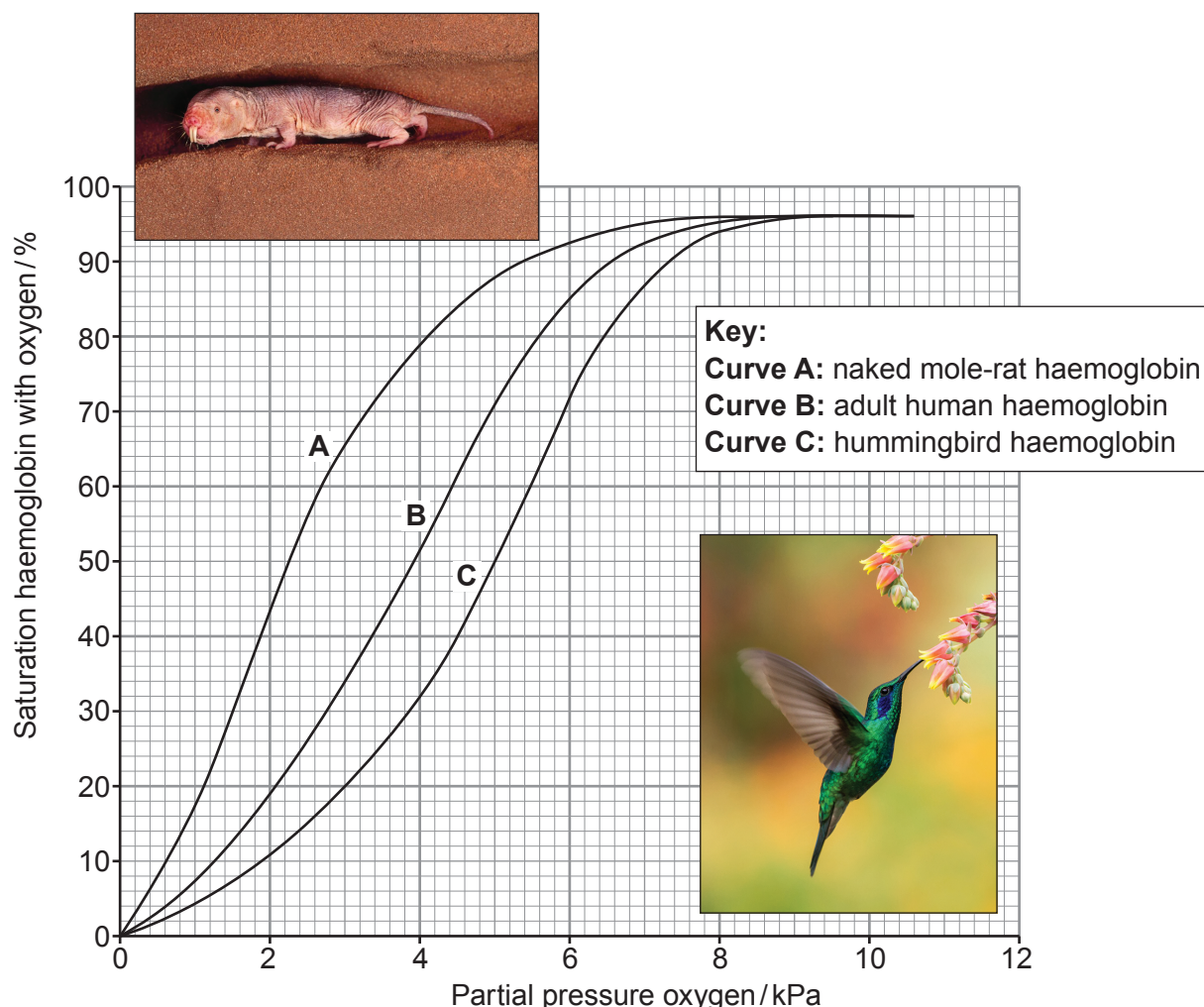


7. **Graph 7** shows the oxygen dissociation curves for the haemoglobin from three different organisms. Haemoglobin **A** is from a naked mole-rat, haemoglobin **B** is from an adult human and haemoglobin **C** is from a hummingbird.

Graph 7



Naked mole-rats live in colonies of up to 80 individuals and spend their entire lives in a maze of poorly ventilated underground tunnels.

Hummingbirds are small birds and their flight muscles account for 30% of their body weight. They can beat their wings up to 200 times per second.

Explain the shape of the oxygen dissociation curve for adult human haemoglobin as shown in **Graph 7**.

Explain the significance of the relative positions of the oxygen dissociation curves for the naked mole-rat haemoglobin and the hummingbird haemoglobin. [9 QER]

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Examiner
only

Area for student response with horizontal ruling lines.



	9

